Fiscal Year:	FY 2010	**Task Last Updated:**	FY 07/30/2009
PI Name:	Judex, Stefan Ph.D.		
Project Title:	Recovery of Musculoskeletal Quantity and Quality upon Multiple Microgravity Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stefan.judex@sunysb.edu	**Fax:**	FY 631-632-8577
PI Organization Type:	UNIVERSITY	**Phone:**	631-632-1549
Organization Name:	SUNY Stony Brook		
PI Address 1:	Dept. of Biomedical Engineering		
PI Address 2:	Bioengineering Building 213		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	10/01/2008	**End Date:**	09/30/2011
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Meck, J@n	**Contact Phone:**	281-244-5405
Contact Email:	janice.v.meck@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rubin, Clinton (SUNY Stony Brook)		
Grant/Contract No.:	NNX08BA35G		
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Task Description:	As space missions become longer, it is apparent that long-term risks of multiple, or even single, exposures to microgravity on crew musculoskeletal health need to be defined and addressed. Even though the magnitude of the long-term deterioration in morphology and strength is largely unknown, it is clear that such risks will be heavily influenced by the genetic make-up of an individual. We have recently identified and characterized a genetically diverse mouse population that, similar to astronauts, includes a diverse range of responses to hindlimb unloading and reambulation. Using this population, here, we will address the consequences of multiple exposures to simulated weightlessness on long-term quantity and quality of trabecular and cortical bone as well as muscle, and investigate how recovery periods between periods of weightlessness can be optimized. Changes in tissue morphology, density, and mechanical properties will be assessed longitudinally in each genetically distinct mouse by in vivo microCT and		

	finite-element modeling, allowing for the identification of skeletal baseline indices that modulate changes in tissue morphology and strength during altered levels of loading. Detailed mechanical tests will be performed at the endpoints to assess the long-term quality of the tissue. Together, the proposed studies provide a unique opportunity to establish the range of responses of genetically diverse individuals to multiple unloading and reambulation phases. Optimizing the length and distribution of recovery periods between disuse exposures may ultimately help to maintain long-term musculoskeletal morphology and strength.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Extended periods of unloading are not confined to space flight but also occur during extended bedrest or paralysis on Earth. Therefore, data from our experiments may help understanding the basis by which the musculoskeleton responds to repeated unloading events and eventually lead to more efficacious phrophylaxes and treatments.
Task Progress:	In the first year of this grant, significant progress has been made towards addressing the specific aims of the proposal. Personnel have been hired as it was proposed in the grant. In Specific Aim 1, it will be determined whether the total exposure time to weightlessness or the number of repetitive exposures is the primary determinant of skeletal morphology and strength. In Specific Aim 2, the musculoskeletal consequences of altering the length and distribution of the recovery periods between multiple exposures to weightlessness will be determined. In experiments concerning these two specific aims, mice in all experimental groups are subject to their first hindlimb unloading (HLU) cycle at 16wk of age, irrespective of the HLU and reambulation (RA) schedule to which they are assigned. However, the age at which they experience additional HLU cycles varies between different experimental groups. Since age alters bone quantity, quality, and remodeling, factors known to influence bone's response to catabolic signals, differences in bone morphology and mechanical strength observed between mice subjected to multiple disuse periods may be attributed not only to differences in their HLU and RA pattern, but to differences in the age at which the subsequent HLU cycles are applied. To address this issue that was also raised by the external reviewers of this grant proposal, a pilot study was initiated to elucidate the effects of aging on disused-induced changes in musculoskeletal quantity and quality, and to confirm the ability of aged mice to survive multiple exposures to HLU. PROGRESS PILOT STUDY – The pilot study has been completed but analyses are ongoing. Forty-four adult (4mo) male B6 mice were assigned to age-matched control (Control) and disuse groups subject to single (D1), double (D2), or triple (D3) (n=11/group) HLU cycles. During each cycle, mice were exposed to continuous HLU for 2wk. Each exposure to HLU was followed by 4wk of reambulation (RA). D3 mice were subjected to 2wk of HLU at 16wk, 22wk, and 28wk of age. To disassociate the concomitant effects of advanced age and prior exposures to HLU on the 22wk and 28 wk HLU cycles for D3 animals, D2 mice were exposed to HLU at 22wk and 28wk, while D1 animals were suspended only at 28wk. All animals were sacrificed at 34wk. Trabecular Bone – At baseline, both the quantity and quality of the trabecular bone of the three experimental disuse groups and the ambulatory controls was statistically similar. All animals, including age-matched controls, experienced substantial loss of trabecular bone between baseline (16wk) and sacrifice (34wk), ranging from 36±5.5% for normal ambulatory control mice to 46±9% for D3 mice. Though the % decrease in bone volume fraction (BV/TV) was the greatest for animals suspended thrice (D3), the difference in total % change of BV/TV was statistically similar for all groups, including ambulatory controls. The loss of bone was accompanied by a significant decrease in the connectivity density and a significant increase in tissue mineral density within groups. As the differences between groups were comparable, the observed changes were primarily driven by age. The relative % decrease in BV/TV during the initial exposure to HLU decreased with increasing age of suspension, though the differences were not statistically significant between groups. On the other hand, the % loss was comparable for all three disuse groups for the HLU cycle from 28wk to 30wk, indicating that prior exposures to HLU were not the primary determinant of the response to subsequent exposures. Both age and previous HLU exposures affected the relative change in BV/TV insomuch as they affected absolute BV/TV values, which were highly correlated with the relative % loss of bone during HLU. Mice in virtually all groups continued to lose trabecular bone even during recovery periods. Analogous to the bone loss experienced during HLU, the losses during recovery periods decreased with increasing age, but were unaffected by the presence of prior RA periods. Losses during recovery were comparable to those experienced by AC, indicating that they too were driven by aging. These findings suggest that the susceptibility of trabecular bone to loss of weight bearing across multiple exposures to disuse and its failure to recover during reloading substantially increases the likelihood of catastrophic failure with each cycle. Metaphyseal Cortical Bone – Mean cortical area (Ct.Ar) of control mice increased over the duration of the study. Unlike trabecular bone in which bone loss continued through the recovery period, Ct.Ar decreased only during HLU cycles, recovering to or surpassing pre-HLU values after each 4wk RA period. However, a significant increase in Ct.Ar from 16wk to 34wk was noted only for the ambulatory controls and D1, indicating that the additional exposures to HLU depressed short-term bone formation within the metaphyseal cortex. The cortex also experienced a similar reduction and recovery of tissue mineralization during HLU cycles and RA periods for most cycles. All groups experienced an increase in TMD from 16wk to 34wk, though a significant difference in the % increase in TMD between groups was only observed between ambulatory controls and D2. However, based on the recovery of both Ct.Ar and TMD during RA periods, longer recovery periods may allow these parameters to normalize to values comparable to ambulatory controls, even for mice exposed to multiple HLU cycles. Diaphyseal Cortical Bone – Analysis of the mid-diaphysis from all in vivo microCT scans is ongoing. PROGRESS SPECIFIC AIM 1 Presently, all one hundred twenty F2 animals, 60 males and 60 females, have been bred and divided between the four experimental groups. All animals have reached skeletally maturity, and baseline microCT scans (16wk age) have been completed. Further, the 1 x 4wk, 2 x 2wk, and 4 x 1wk disuse groups have recently been subjected to and completed a single HLU cycle of 28d, 14d, and 7d, respectively. Based on the initial HLU cycle, and results from the pilot study, we anticipate that the 2 x 2wk and the 4 x 1wk disuse animals will successfully sustain subsequent HLU cycles. Preliminary ¬in-vivo microCT results indicate that the F2 mice population exhibits heterogeneous baseline skeletal morphology as

	expected. 3-D evaluation of both the distal metaphysis and mid-diaphysis at baseline and after a single HLU cycle is presently underway for all experimental groups. PROGRESS SPECIFIC AIM 2 One hundred twenty F2 mice are currently being bred for the experiments proposed in SA2.
Bibliography Type:	Description: (Last Updated: 01/16/2018)
Abstracts for Journals and Proceedings	Gupta S, Uzer G, Judex S. "Bone Atrophy and Recovery upon Multiple Exposures to Mechanical Unloading." To be presented at the 31st Annual Meeting of the American Society of Bone and Mineral Research, Denver, CO, September 2009. Journal of Bone and Mineral Research, abstract issue, September 2009. , Sep-2009
Articles in Peer-reviewed Journals	Judex S, Gupta S, Rubin C. "Regulation of mechanical signals in bone." Orthodontics and Craniofacial Research 2009 May;12(2):94-104. PubMed PMID: 19419452 , May-2009
Articles in Peer-reviewed Journals	Bandstra ER, Thompson RW, Nelson GA, Willey JS, Judex S, Cairns MA, Benton ER, Vazquez ME, Carson JA, Bateman TA. "Musculoskeletal changes in mice from 20-50 cGy of simulated galactic cosmic rays." Radiat Res. 2009 Jul;172(1):21-9. PubMed PMID: 19580504 , Jul-2009